

Renormalization in Massive ϕ^4 Theory

Source: Perimeter Scholars International, Quantum Field Theory II, Homework 2; 15 worked parts.

CONTENTS

- 1 Renormalization for Massive ϕ^4 Theory
- 2 Renormalization Group Flow for the $O(N)$ Model

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1 Renormalization for Massive ϕ^4 Theory

Work in Euclidean signature and retain terms through one loop. Define the renormalized action by

$$S_R[\phi] = S[\phi] + \hbar \Delta_1 S[\phi] + O(\hbar^2), \quad (1.1)$$

where

$$S[\phi] = \int d^4x \left[\frac{1}{2}(\partial\phi)^2 + \frac{m_R^2}{2}\phi^2 + \frac{g_R}{4!}\phi^4 \right] \quad (1.2)$$

and

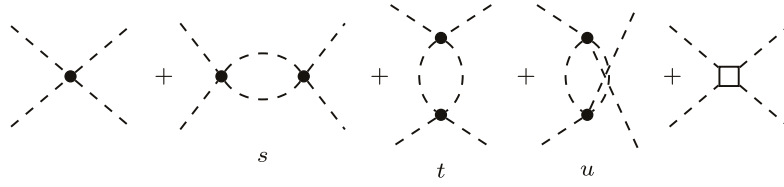
$$\Delta_1 S[\phi] = \int d^4x \left[\frac{B_1}{2}\phi^2 + \frac{C_1}{4!}\phi^4 \right]. \quad (1.3)$$

The cutoff-dependent coefficients B_1 and C_1 are chosen so that physical quantities expressed in terms of m_R and g_R remain finite as $\Lambda \rightarrow \infty$.

The one-loop four-point 1PI function is

$$\begin{aligned} \hat{\Gamma}_R^{(4)}(p_i) = & g_R - \frac{\hbar g_R^2}{2} \left[I(p_1 + p_2, m_R) + I(p_1 + p_3, m_R) + I(p_1 + p_4, m_R) \right] \\ & + \hbar C_1 + O(\hbar^2). \end{aligned} \quad (1.4)$$

Using a hollow square for the quartic counterterm, these contributions have the following topologies:



1. In four-dimensional Euclidean spacetime, write the sharp-cutoff integral corresponding to

$$I(p_1 + p_2, m_R; \Lambda), \quad (1.5)$$

where Λ regulates the ultraviolet divergence.

Solution.

$$\begin{aligned} I(p_1 + p_2, m_R; \Lambda) &= \int \frac{d^4 k}{(2\pi)^4} \frac{1}{(k^2 + m_R^2)[(k + p_1 + p_2)^2 + m_R^2]} \\ &= \frac{4\pi}{(2\pi)^4} \int_0^\pi d\chi \sin^2 \chi \int_0^\Lambda dk \frac{k^3}{(k^2 + m_R^2)(k^2 + s + 2k\sqrt{s} \cos \chi + m_R^2)}. \end{aligned} \quad (1.6)$$

2. Set $m_R^2 = 0$ and calculate the leading large- Λ behavior of $I(p, 0; \Lambda)$.

Solution.

$$\begin{aligned} I(p_1 + p_2, 0; \Lambda) &= \frac{1}{4\pi^3} \int_0^\Lambda dk k \int_0^\pi d\chi \frac{\sin^2 \chi}{k^2 + s + 2k\sqrt{s} \cos \chi} \\ &= \frac{1}{4\pi^3} \int_0^\Lambda dk k \frac{\pi}{4k^2 s} \left(k^2 + s - \sqrt{(k^2 + s)^2 - 4k^2 s} \right) \\ &= \frac{1}{16\pi^2 s} \int_0^\Lambda \frac{1}{k} (k^2 + s - |k^2 - s|) \\ &= \frac{1}{16\pi^2 s} \left(\int_{\sqrt{s}}^\Lambda dk \frac{2s}{k} + \int_0^{\sqrt{s}} dk \frac{2k^2}{k} \right) \\ &= \frac{1}{16\pi^2 s} \left(1 + \log \frac{\Lambda^2}{s} \right). \end{aligned} \quad (1.7)$$

3. By differentiating under the integral, show that

$$\frac{\partial}{\partial(m_R^2)} I(p, m_R; \Lambda) \quad (1.8)$$

is finite as $\Lambda \rightarrow \infty$. Conclude that the divergent part in the massive theory is the same as the one identified in the massless theory.

Solution. By differentiating with respect to m_R^2 , the integral now looks like

$$\sim \int_0^\Lambda dk \frac{k^3}{k^6}, \quad (1.9)$$

which is finite. Since the rate of change of the integral remains finite, the divergent part must be the same for massive and massless theory.

For parts d)--h), work in massless ϕ^4 theory.

4. Determine the divergent part of C_1 required to make $\widehat{\Gamma}_R^{(4)}(p_i)$ finite and impose

$$\widehat{\Gamma}_R^{(4)}(p_i^{\text{ref}}) = g_R, \quad |p_1^{\text{ref}} + p_2^{\text{ref}}| = |p_1^{\text{ref}} + p_3^{\text{ref}}| = |p_1^{\text{ref}} + p_4^{\text{ref}}| = \mu. \quad (1.10)$$

Only the terms in the counterterm that diverge as $\Lambda \rightarrow \infty$ need be given explicitly.

Solution. Using the result from b), we have

$$g_R - \frac{\hbar g_R^2}{2} \frac{3}{16\pi^2} \left(1 + \log \frac{\Lambda^2}{\mu^2}\right) + \hbar C_1 = g_R \implies C_1 = \frac{3g_R^2}{32\pi^2} \left(1 + \log \frac{\Lambda^2}{\mu^2}\right). \quad (1.11)$$

5. Find $\widehat{\Gamma}_R^{(4)}(p_i, g_R, \mu)$ at general external momenta. Renormalization-scale independence requires

$$\widehat{\Gamma}_R^{(4)}(p_i, g_R, \mu) = \widehat{\Gamma}_R^{(4)}(p_i, g'_R, \mu'), \quad (1.12)$$

where the primed parameters are defined at the scale μ' .

Solution.

$$\begin{aligned} \widehat{\Gamma}_R^{(4)}(p_i, g_R, \mu) &= g_R - \frac{\hbar g_R^2}{32\pi^2} \left(\log \frac{\mu^2}{s} + \log \frac{\mu^2}{t} + \log \frac{\mu^2}{u} \right) \\ &= g'_R - \frac{\hbar g_R'^2}{32\pi^2} \left(\log \frac{\mu'^2}{s} + \log \frac{\mu'^2}{t} + \log \frac{\mu'^2}{u} \right). \end{aligned} \quad (1.13)$$

6. Evaluate the preceding equality at a convenient momentum configuration and derive $g'_R(\mu')$ in terms of $g_R(\mu)$, μ , and μ' .

Solution. Take $s = u = t = \mu'^2$, we find

$$g'_R(\mu') = g_R(\mu) - \frac{3\hbar g_R^2(\mu)}{32\pi^2} \log \frac{\mu^2}{\mu'^2}. \quad (1.14)$$

7. Define the effective coupling by

$$g_{\text{eff}}(E) = \begin{cases} g_R(\mu), & E = \mu, \\ g'_R(\mu'), & E = \mu', \end{cases} \quad (1.15)$$

and the beta function by

$$\beta(g_{\text{eff}}) \equiv E \frac{d}{dE} g_{\text{eff}}(E). \quad (1.16)$$

Compute the one-loop beta function and plot it as a function of g_{eff} .

Solution. The definition here is not very conventional. It is equivalent to defining

$$g_{\text{eff}}(E) = \Gamma_R^{(4)}(E, g_R, \mu), \quad (1.17)$$

which immediately implies

$$g_{\text{eff}}(E) = \begin{cases} g_R(\mu), & E = \mu, \\ g'_R(\mu'), & E = \mu'. \end{cases} \quad (1.18)$$

Then, we have

$$\beta(g_{\text{eff}}) = \frac{3\hbar g_R^2(\mu)}{16\pi^2} = \frac{3\hbar g_{\text{eff}}^2}{16\pi^2}. \quad (1.19)$$

8. Explain what this beta function implies for the interaction strength at high and low energies.

Solution. Solving the beta function, we have

$$\frac{1}{g_{\text{eff}}(E)} = \frac{1}{g_0} + \frac{3\hbar}{32\pi^2} \log \frac{\Lambda^2}{E^2}. \quad (1.20)$$

Therefore, the theory is free at IR and becomes strongly coupled at the UV.

The four-point diagrams introduce no additional divergence in the massive theory, so the same calculation can be repeated there. For mass renormalization, consider the one-loop two-point 1PI function

$$\hat{\Gamma}_R^{(2)}(p) = p^2 + m_R^2 + \frac{\hbar g_R}{2} T(m_R) + \hbar B + O(\hbar^2). \quad (1.21)$$

Using a filled square for the quadratic counterterm, its topologies are



1. Write the sharp-cutoff integral $T(m_R; \Lambda)$ that regulates the UV divergences.

Solution.

$$T(m_R; \Lambda) = \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m^2} = \frac{1}{8\pi^2} \int_0^\Lambda dk \frac{k^3}{k^2 + m^2}. \quad (1.22)$$

2. Evaluate $T(m_R; \Lambda)$ at large Λ . Identify separately the quadratic divergence, logarithmic divergence, and term that remains finite as $\Lambda \rightarrow \infty$.

Hint: Radial coordinates may be useful. The area of the unit n -sphere is

$$\int d^n \Omega = \frac{2\pi^{n/2}}{\Gamma(n/2)}. \quad (1.23)$$

Solution.

$$T(m_R; \Lambda) = \frac{1}{16\pi^2} \left(\Lambda^2 - m_R^2 \log \frac{\Lambda^2 + m_R^2}{m_R^2} \right) = \frac{1}{16\pi^2} \left(\Lambda^2 - m_R^2 \log \frac{\Lambda^2}{m_R^2} + \mathcal{O}(\Lambda^{-2}) \right). \quad (1.24)$$

The physical mass is determined by the pole of the propagator, equivalently by

$$\hat{\Gamma}_R^{(2)}(p_i^{\text{ref}}, m_R, g_R, \mu) = 0, \quad |p_i^{\text{ref}}| = m_{\text{phys}}, \quad (1.25)$$

for any μ . For practical subtraction, impose instead

$$\hat{\Gamma}_R^{(2)}(0, m_R, g_R, \mu) = m_R^2, \quad \mu = m_R, \quad (1.26)$$

and take the mass counterterm to have the form

$$B(m_R, g_R, \mu, \Lambda) = B_0(g_R, \mu, \Lambda) + m_R^2 B_1(g_R, \mu, \Lambda). \quad (1.27)$$

3. Determine $B_0(g_R, \mu, \Lambda)$ and $B_1(g_R, \mu, \Lambda)$ so that the preceding renormalization condition holds.

Solution. We can rewrite

$$\hat{\Gamma}_R^{(2)}(p, m_R, g_R, \mu) = p^2 + m_R^2 + \hbar \left(\frac{g_R(\mu)\Lambda^2}{32\pi^2} + B_0 + m_R^2 \left(-\frac{g_R(\mu)}{32\pi^2} \log \frac{\Lambda^2}{\mu^2} - \frac{g_R(\mu)}{32\pi^2} \log \frac{\mu^2}{m_R^2} + B_1 \right) \right). \quad (1.28)$$

The renormalisation condition implies

$$B_1(g_R, \mu, \Lambda) = \frac{g_R}{32\pi^2} \log \frac{\Lambda^2}{\mu^2}, \quad B_0(g_R, \mu, \Lambda) = -\frac{g_R \Lambda^2}{32\pi^2}. \quad (1.29)$$

4. Check that these counterterms make $\hat{\Gamma}_R^{(2)}(p_i, m_R, g_R, \mu)$ finite for arbitrary m_R and p .

Solution.

$$\hat{\Gamma}_R^{(2)}(p, m_R, g_R, \mu) = p^2 + m_R(\mu)^2 - \frac{\hbar g_R(\mu)}{32\pi^2} \log \frac{\mu^2}{m_R(\mu)^2}. \quad (1.30)$$

5. Require $\hat{\Gamma}_R^{(2)}$ to be independent of the renormalization scale, as for the four-point function, and derive an equation relating $[m'_R(\mu')]^2$ and $[m_R(\mu)]^2$. Retain terms only through order $\hbar$ and express the result in terms of g_R .

Solution.

$$m_R^2(\mu) - \frac{\hbar g_R(\mu) m_R^2(\mu)}{32\pi^2} \log \left(\frac{\mu^2}{m_R^2(\mu)} \right) = [m'_R(\mu')]^2 - \frac{\hbar g'_R(\mu') [m'_R(\mu')]^2}{32\pi^2} \log \left(\frac{\mu'^2}{[m'_R(\mu')]^2} \right). \quad (1.31)$$

This implies

$$[m'_R(\mu')]^2 = m_R^2(\mu) + \frac{\hbar g_R(\mu) m_R^2(\mu)}{32\pi^2} \log\left(\frac{\mu'^2}{\mu^2}\right) + O(\hbar^2). \quad (1.32)$$

6. Show that when $\mu^2 = m_R^2$, the renormalized mass agrees in magnitude with the physical mass:

$$|m_R| = |m_{\text{phys}}|. \quad (1.33)$$

Solution.

$$\hat{\Gamma}_R^{(2)}(|m_{\text{phys}}|, m_R = \mu, g_R, \mu) = 0 \implies |m_{\text{phys}}| = |m_R|. \quad (1.34)$$

7. Calculate the anomalous dimension of the mass-squared operator,

$$\gamma_{m^2}(g_R) \equiv \mu \frac{\partial}{\partial \mu} \ln m_R^2(\mu). \quad (1.35)$$

Solution. Starting from the physical mass

$$m_{\text{phys}}^2 = m_R^2(\mu) \left[1 - \frac{\hbar g_R(\mu)}{32\pi^2} \log\left(\frac{\mu^2}{m_R^2(\mu)}\right) \right] + O(\hbar^2), \quad (1.36)$$

and hence

$$\log m_{\text{phys}}^2 = \log m_R^2(\mu) - \frac{\hbar g_R(\mu)}{32\pi^2} \log\left(\frac{\mu^2}{m_R^2(\mu)}\right) + O(\hbar^2). \quad (1.37)$$

Then,

$$0 = \frac{d \log m_{\text{phys}}^2}{d \log \mu} = \gamma_{m^2}(g_R) - \frac{2\hbar g_R}{32\pi^2} + O(\hbar^2), \quad (1.38)$$

which implies

$$\gamma_{m^2}(g_R) = \frac{\hbar g_R}{16\pi^2} + O(\hbar^2). \quad (1.39)$$

At one loop in ϕ^4 theory, no wavefunction renormalization is required; the two-point correction is momentum independent at this order.

2 Renormalization Group Flow for the $O(N)$ Model

Let $\vec{\phi} = \{\phi_\alpha\}_{\alpha=1}^N$ be an N -component scalar field with $O(N)$ symmetry. In D -dimensional Euclidean space, take

$$S_E[\vec{\phi}] = \int d^D x \left[\frac{1}{2} (\nabla \vec{\phi})^2 + U(\vec{\phi}^2) \right], \quad (2.1)$$

where

$$X \equiv \vec{\phi}^2 = \sum_{\alpha=1}^N \phi_\alpha^2, \quad (2.2)$$

and $U(X)$ is regular. Restrict the Fourier support to $|k| \leq \Lambda$ and write

$$Z = \int \mathcal{D}_\Lambda[\vec{\phi}] \exp\left[-\frac{1}{\hbar} S_E[\vec{\phi}]\right]. \quad (2.3)$$

Separate the field into slow and fast components,

$$\vec{\phi} = \tilde{\vec{\phi}} + \vec{\phi}_{\text{fast}}, \quad (2.4)$$

where $\tilde{\vec{\phi}}$ has modes $|k| \leq \Lambda/S$ and $\vec{\phi}_{\text{fast}}$ has modes $\Lambda/S < |k| \leq \Lambda$. Integrate a thin shell using

$$S = 1 + ds, \quad 0 < ds \ll 1. \quad (2.5)$$

1. Hold $\tilde{\vec{\phi}}$ fixed and expand $U(X)$ through second order in $\vec{\phi}_{\text{fast}}$ (the Gaussian approximation), expressing the result in terms of U' and U'' . Keep the vector structure explicit.
2. In the local-potential approximation, take $\tilde{\vec{\phi}}$ to be constant, so only its $k = 0$ mode is nonzero. Show that the fast-mode integral separates into one longitudinal mode parallel to $\tilde{\vec{\phi}}$ and $N - 1$ decoupled transverse modes.
3. Show that the two kinds of mode renormalize the potential according to

$$U \longrightarrow U_{\text{eff}} = U + ds (\Delta U_{\parallel} + (N - 1)\Delta U_{\perp}), \quad (2.6)$$

and calculate $\Delta U_{\parallel}(X)$ and $\Delta U_{\perp}(X)$.

4. Show that the rescalings

$$\vec{\phi} \longrightarrow (1 - ds)^{(2-D)/2} \vec{\phi}, \quad x \longrightarrow (1 - ds)x \quad (2.7)$$

restore the cutoff to Λ .

5. Determine the local-potential flow equation

$$S \frac{\partial}{\partial S} U_S(X) = ? \quad (2.8)$$

for the $O(N)$ model. As in the lecture, set $\Lambda = 1$ if desired.

6. Check that $N = 1$ reproduces the one-component flow for $V(\phi)$, using $U(\phi^2) = V(\phi)$.
7. Truncate the potential at second order in $X = \vec{\phi}^2$,

$$U(X) = \frac{t}{2} X + \frac{u}{8} X^2. \quad (2.9)$$

Expand the flow equation consistently through X^2 . Derive both the flow equations and the Wilson functions $S \partial g_S / \partial S$ for the renormalized couplings t_S and u_S .